Ali Rezaei

Born: 18th April 1998
University: University of Ljubljana
Department: Mechanical Engineering
Marital Status: Single

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Profile

I am a Ph.D. researcher in Mechanical Engineering at the University of Ljubljana and a member of the LASTEH (Laboratory for laser techniques) research group. My research focuses on low-dimensional-based nanomaterials, laser ultrasonics, photoacoustics, and nanomaterial-based piezophotonic composites. I have experience in both experimental and computational research, scientific publishing, and interdisciplinary collaboration, with strong skills in numerical modeling, scientific programming, and optical measurement techniques.

Education

Ph.D. 2024 – Present

University of Ljubljana, Slovenia

Mechanical Engineering

> Two-dimensional materials-based piezophotonic composites for laser ultrasound generation

Master's degree 2020 – 2023

University of Isfahan, Iran

Optics-Laser Physics

> Investigation of Klein tunneling effect in the presence of tilted Dirac cones

Bachelor's degree 2016 – 2020

University of Isfahan

Physics

Publications

Ultrafast subwavelength CVD-graphene nanoheater for the generation of broadband photoacoustic waves, Ali Rezaei, Diogo A Pereira, Giuseppe Valerio Bianco, Giovanni Bruno, Aleš Mrzel, LG Arnaut, Carlos Serpa, Matija Jezeršek, and Daniele Vella, Photoacoustics Volume 47, 2026, 100796 https://doi.org/10.1016/j.pacs.2026.100796 — paper

Efficient decoration of graphene oxide with a narrow size distribution of noble metal nanoparticles: Green reduction and integration into a thermoelastic composite, Daniele Vella, Damjan Vengust, Polona Umek, Ali Rezaei, Matija Jezeršek, and Aleš Mrzel, Synthetic Metals 313 (2025) 117894

10.1016/j.synthmet.2025.117894 — paper

Tilted Dirac cones in two-dimensional materials: Impact on electron transmission and pseudospin dynamics, Rasha Al-Marzoog, Ali Rezaei, Zahra Noorinejad, Mohsen Amini, Ebrahim Ghanbari-Adivi, and Seyed Akbar Jafari, *Phys. Rev. B 110, 165427* https://doi.org/10.1103/PhysRevB.110.165427 — paper

Large-area graphene–PDMS interface: ultrathin nano-heaters for photoacoustic wave generation via picosecond laser excitation, Ali Rezaei, Diogo A. Pereira, Giuseppe Valerio Bianco, Giovanni Bruno, Aleš — conference paper

Mrzel, Luís G. Arnaut, Carlos Serpa, Matija Jezeršek, and Daniele Vella, *Book of Abstracts of Progress in Photoacoustic & Photothermal Phenomena*, Erice, Sicily, Italy, 2025, p. 60.
https://plus-legacy.cobiss.net/cobiss/si/sl/bib/257944323

Teaching Experience

Physics I, Physics II, Classical Mechanics, and General physics. 2022 – 2023

Teacher Assistant (Undergraduate Course), University of Isfahan

Research Interests

- *Photoacoustics and Laser Ultrasonics*
- *Graphene and Two-Dimensional (2D) Materials*
- *Piezophotonic and Thermoelastic Composites*
- *Ultrafast Laser–Matter Interaction*
- *Optics and Laser Physics*
- *Wave Propagation and Non-destructive Testing (NDT)*
- *Nanomaterials and Functional Composites*
- *Computational Modelling and Multiphysics Simulation (COMSOL, MATLAB, Python)*

Skills

Teaching	Physics Mathematics English language
Programming	Python: Professional with valid certificate (packages: NumPy, Pandas, matplotlib, SciPy, SymPy, Scrapy, Plotly, mpi4py) MATLAB: Professional
Others	COMSOL Microsoft office tools LATEX ICDL Adobe Photoshop Vesta SolidWorks

Languages

Persian	Native
English	Fluent in speaking and proficient in writing **IELTS**, Academic (2023-09-30), **Overall Band Score: 6.5** (L:7.0, R:6.0, W:6.0, S:6.0)
Arabic	Fair